

## Projet UL2IN002 – 2026fev Thème : “la Mer”

|                        |
|------------------------|
| Numéro de groupe : TBD |
|------------------------|

|                        |               |
|------------------------|---------------|
| Nom : Benkelfat        | Nom :         |
| Prénom : Imad          | Prénom :      |
| N° étudiant : 21408644 | N° étudiant : |

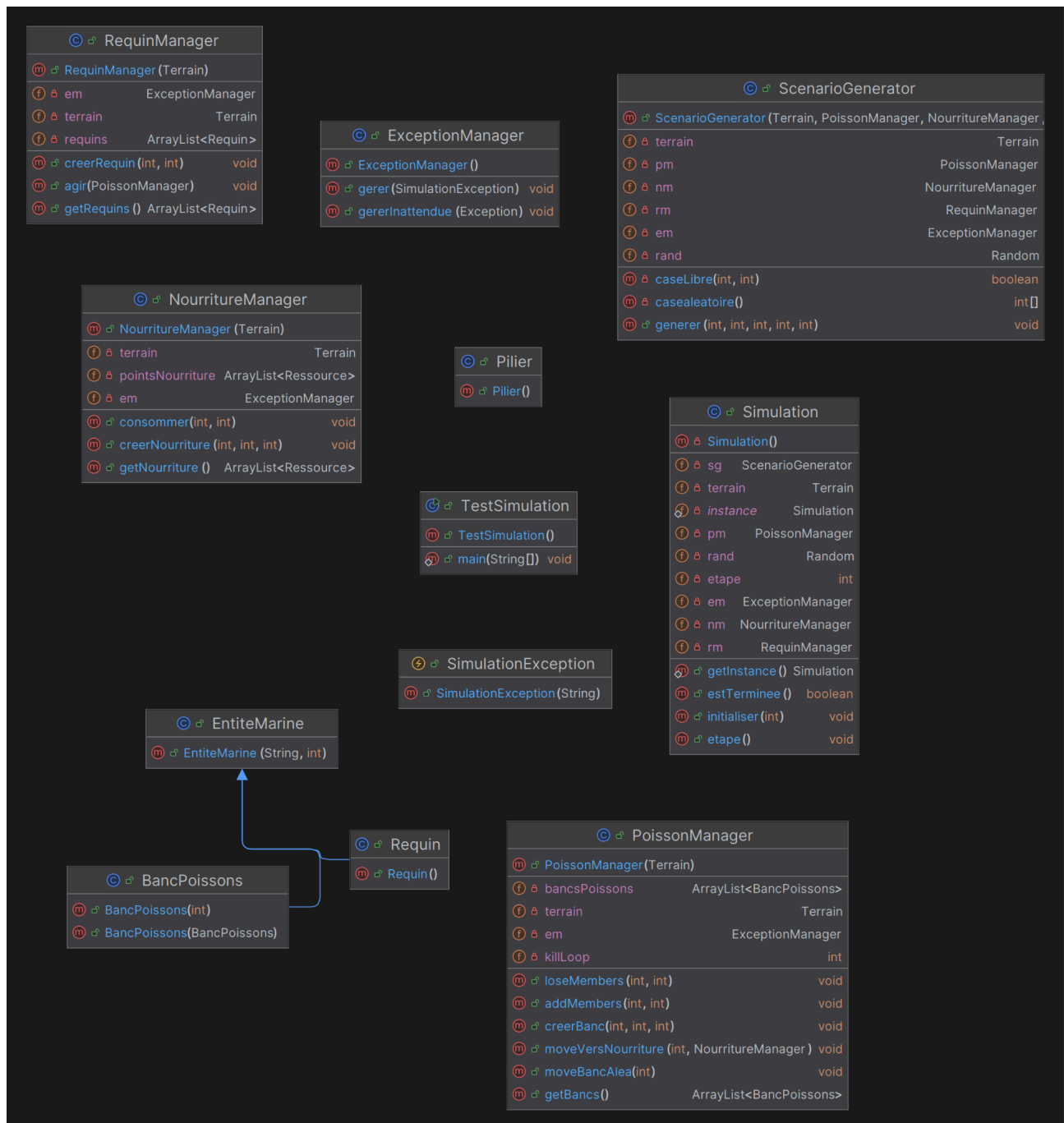
|                                                                          | <i><b>Thème de la simulation</b></i>                           |
|--------------------------------------------------------------------------|----------------------------------------------------------------|
| <i>Thème et objectif(s) de votre simulation :<br/>(5 lignes maximum)</i> | Simulation d'un océan avec prédateurs et sources de nourriture |
| <i>Les agents :</i>                                                      | Requins, Bancs de poissons                                     |
| <i>Les ressources :</i>                                                  | Piliers, Nourriture                                            |

|                                                                                                                  |     |
|------------------------------------------------------------------------------------------------------------------|-----|
| Quel est le nombre de classes/interfaces de votre projet ?<br>(ne pas compter Ressource, Terrain et TestTerrain) | 12  |
| Avez-vous une hiérarchie d'héritage à 3 niveaux ?                                                                | Oui |
| Avez-vous généré la javadoc ?                                                                                    | Oui |
| Avez-vous généré des logs ?                                                                                      | Oui |

| <i><b>Checklist des points vus en cours à mettre en application :</b></i> | <i><b>Nom de classe correspondant (une seule classe demandée, mais vous pouvez en mettre plus)</b></i> |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Classe contenant une ArrayList                                            | PoissonsManager                                                                                        |
| Classe abstraite et méthode abstraite                                     | EntitéMarine                                                                                           |
| Interface                                                                 | RequinManager                                                                                          |
| Classe avec un constructeur par copie ou méthode clone()                  | BancPoissons                                                                                           |
| Classe contenant que des attributs et méthodes statiques                  | SimulationConfig                                                                                       |
| Classe héritant de Exception                                              | SimulationException                                                                                    |
| Gestion des exceptions                                                    | ExceptionManager                                                                                       |

|                                   |            |
|-----------------------------------|------------|
| Singleton ou une de ses variantes | Simulation |
|-----------------------------------|------------|

*Schéma UML des classes vision fournisseur (dessin “à la main” scanné ou photo acceptés)*



*Copier-coller de 2 ou 3 logs (affichages de simulations)*

Scénario généré : 3 bancs, 5 nourritures, 3 requins, 2 piliers

=== Etape 1 ===





|        |        |            |            |            |  |        |            |         |  |  |  |
|--------|--------|------------|------------|------------|--|--------|------------|---------|--|--|--|
| Requin | Requin |            |            |            |  |        |            |         |  |  |  |
|        |        |            |            |            |  |        |            |         |  |  |  |
|        |        | Nourriture |            |            |  |        | Nourriture |         |  |  |  |
|        |        |            |            |            |  | Pilier |            |         |  |  |  |
|        |        |            | Requin     |            |  |        |            |         |  |  |  |
|        |        |            |            |            |  |        | Pilier     | Poisson |  |  |  |
|        |        |            | Nourriture |            |  |        | Poisson    |         |  |  |  |
|        |        |            |            |            |  |        |            |         |  |  |  |
|        |        |            | Poisson    | Nourriture |  |        |            |         |  |  |  |

Poisson mange près de (10,4)  
Bancs : 3 | Nourritures : 4

=== Etape 4 ===

|        |        |            |            |  |  |        |            |         |  |  |  |
|--------|--------|------------|------------|--|--|--------|------------|---------|--|--|--|
|        |        |            |            |  |  |        |            |         |  |  |  |
| Requin | Requin |            |            |  |  |        |            |         |  |  |  |
|        |        |            |            |  |  |        |            |         |  |  |  |
|        |        | Nourriture |            |  |  |        | Nourriture |         |  |  |  |
|        |        |            |            |  |  | Pilier |            |         |  |  |  |
|        |        |            | Requin     |  |  |        |            | Poisson |  |  |  |
|        |        |            |            |  |  |        | Pilier     |         |  |  |  |
|        |        |            | Nourriture |  |  |        | Poisson    |         |  |  |  |
|        |        |            |            |  |  |        |            |         |  |  |  |
|        |        |            |            |  |  |        |            |         |  |  |  |

Poisson mange près de (8,4)  
Poisson mange près de (8,4)  
Poisson mange près de (4,8)  
Bancs : 3 | Nourritures : 3

=== Etape 5 ===

|        |        |            |         |         |  |        |            |  |  |  |
|--------|--------|------------|---------|---------|--|--------|------------|--|--|--|
|        |        |            |         |         |  |        |            |  |  |  |
| Requin | Requin |            |         |         |  |        |            |  |  |  |
|        |        |            |         |         |  |        |            |  |  |  |
|        |        | Nourriture |         |         |  |        | Nourriture |  |  |  |
|        |        |            |         |         |  | Pilier | Poisson    |  |  |  |
|        |        |            | Requin  |         |  |        |            |  |  |  |
|        |        |            |         |         |  |        | Pilier     |  |  |  |
|        |        |            |         | Poisson |  |        |            |  |  |  |
|        |        |            | Poisson |         |  |        |            |  |  |  |
|        |        |            |         |         |  |        |            |  |  |  |

[REQUIN] mange un poisson du banc en (7,4)  
[REQUIN] mange un poisson du banc en (-1,-1)  
Poisson mange près de (4,8)  
Bancs : 2 | Nourritures : 3

### === Etape 6 ===

|        |        |            |        |  |  |        |            |  |  |  |
|--------|--------|------------|--------|--|--|--------|------------|--|--|--|
|        |        |            |        |  |  |        |            |  |  |  |
| Requin | Requin |            |        |  |  |        |            |  |  |  |
|        |        |            |        |  |  |        |            |  |  |  |
|        |        | Nourriture |        |  |  |        | Nourriture |  |  |  |
|        |        |            |        |  |  | Pilier |            |  |  |  |
|        |        |            | Requin |  |  |        |            |  |  |  |
|        |        |            |        |  |  |        | Pilier     |  |  |  |
|        |        | Poisson    |        |  |  |        |            |  |  |  |
|        |        |            |        |  |  |        |            |  |  |  |
|        |        |            |        |  |  |        |            |  |  |  |

[REQUIN] mange un poisson du banc en (7,3)  
[REQUIN] mange un poisson du banc en (-1,-1)

Bancs : 1 | Nourritures : 3

=== Etape 7 ===

|        |        |            |        |  |  |        |         |  |  |  |  |  |
|--------|--------|------------|--------|--|--|--------|---------|--|--|--|--|--|
|        |        |            |        |  |  |        |         |  |  |  |  |  |
| Requin | Requin |            |        |  |  |        |         |  |  |  |  |  |
|        |        |            |        |  |  |        | Poisson |  |  |  |  |  |
|        |        | Nourriture |        |  |  |        |         |  |  |  |  |  |
|        |        |            |        |  |  | Pilier |         |  |  |  |  |  |
|        |        |            | Requin |  |  |        |         |  |  |  |  |  |
|        |        |            |        |  |  |        | Pilier  |  |  |  |  |  |
|        |        |            |        |  |  |        |         |  |  |  |  |  |
|        |        |            |        |  |  |        |         |  |  |  |  |  |
|        |        |            |        |  |  |        |         |  |  |  |  |  |
|        |        |            |        |  |  |        |         |  |  |  |  |  |

Bancs : 1 | Nourritures : 3

=== Etape 8 ===

|        |        |            |        |  |  |         |        |  |  |  |  |  |
|--------|--------|------------|--------|--|--|---------|--------|--|--|--|--|--|
|        |        |            |        |  |  |         |        |  |  |  |  |  |
| Requin | Requin |            |        |  |  |         |        |  |  |  |  |  |
|        |        |            |        |  |  |         |        |  |  |  |  |  |
|        |        | Nourriture |        |  |  | Poisson |        |  |  |  |  |  |
|        |        |            |        |  |  | Pilier  |        |  |  |  |  |  |
|        |        |            | Requin |  |  |         |        |  |  |  |  |  |
|        |        |            |        |  |  |         | Pilier |  |  |  |  |  |
|        |        |            |        |  |  |         |        |  |  |  |  |  |
|        |        |            |        |  |  |         |        |  |  |  |  |  |
|        |        |            |        |  |  |         |        |  |  |  |  |  |
|        |        |            |        |  |  |         |        |  |  |  |  |  |

Bancs : 1 | Nourritures : 3

=== Etape 9 ===

|        |        |            |        |  |         |        |  |  |  |  |  |  |
|--------|--------|------------|--------|--|---------|--------|--|--|--|--|--|--|
|        |        |            |        |  |         |        |  |  |  |  |  |  |
|        |        |            |        |  |         |        |  |  |  |  |  |  |
| Requin | Requin |            |        |  |         |        |  |  |  |  |  |  |
|        |        |            |        |  |         |        |  |  |  |  |  |  |
|        |        |            |        |  |         |        |  |  |  |  |  |  |
|        |        | Nourriture |        |  | Poisson |        |  |  |  |  |  |  |
|        |        |            |        |  | Pilier  |        |  |  |  |  |  |  |
|        |        |            | Requin |  |         |        |  |  |  |  |  |  |
|        |        |            |        |  |         | Pilier |  |  |  |  |  |  |
|        |        |            |        |  |         |        |  |  |  |  |  |  |
|        |        |            |        |  |         |        |  |  |  |  |  |  |
|        |        |            |        |  |         |        |  |  |  |  |  |  |
|        |        |            |        |  |         |        |  |  |  |  |  |  |
|        |        |            |        |  |         |        |  |  |  |  |  |  |

Bancs : 1 | Nourritures : 3

=== Etape 10 ===

|        |        |            |        |         |        |        |  |  |  |  |  |  |
|--------|--------|------------|--------|---------|--------|--------|--|--|--|--|--|--|
|        |        |            |        |         |        |        |  |  |  |  |  |  |
|        |        |            |        |         |        |        |  |  |  |  |  |  |
| Requin | Requin |            |        |         |        |        |  |  |  |  |  |  |
|        |        |            |        |         |        |        |  |  |  |  |  |  |
|        |        |            |        |         |        |        |  |  |  |  |  |  |
|        |        | Nourriture |        | Poisson |        |        |  |  |  |  |  |  |
|        |        |            |        |         | Pilier |        |  |  |  |  |  |  |
|        |        |            | Requin |         |        |        |  |  |  |  |  |  |
|        |        |            |        |         |        | Pilier |  |  |  |  |  |  |
|        |        |            |        |         |        |        |  |  |  |  |  |  |
|        |        |            |        |         |        |        |  |  |  |  |  |  |
|        |        |            |        |         |        |        |  |  |  |  |  |  |
|        |        |            |        |         |        |        |  |  |  |  |  |  |

Poisson mange près de (4,3)

Bancs : 1 | Nourritures : 3

=== Etape 11 ===



|        |        |            |         |  |  |        |        |  |  |  |  |
|--------|--------|------------|---------|--|--|--------|--------|--|--|--|--|
|        |        |            |         |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  |        |        |  |  |  |  |
| Requin | Requin |            |         |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  |        |        |  |  |  |  |
|        |        | Nourriture | Poisson |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  | Pilier |        |  |  |  |  |
|        |        |            | Requin  |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  |        | Pilier |  |  |  |  |
|        |        |            |         |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  |        |        |  |  |  |  |
|        |        |            |         |  |  |        |        |  |  |  |  |

Bancs : 0 | Nourritures : 3

=== Simulation terminée ===

PS C:\Users\Imad\Documents\GitHub\Cours-Sorbonne-Imad\S4\Java\Projet  
Java\UL2IN002-PROJET\code\Imad\Nouveau dossier (2)>java TestSimulation 0  
Scénario généré : 3 bancs, 2 nourritures, 2 requins, 2 piliers

=== Etape 1 ===

|            |  |  |            |         |         |        |  |        |  |        |  |
|------------|--|--|------------|---------|---------|--------|--|--------|--|--------|--|
|            |  |  |            |         |         |        |  |        |  |        |  |
| Nourriture |  |  |            |         |         |        |  |        |  |        |  |
|            |  |  |            |         |         |        |  |        |  | Requin |  |
|            |  |  |            |         |         |        |  |        |  |        |  |
|            |  |  |            |         |         |        |  |        |  |        |  |
|            |  |  | Nourriture |         |         | Requin |  |        |  |        |  |
|            |  |  |            |         |         |        |  |        |  |        |  |
|            |  |  | Poisson    |         |         |        |  |        |  |        |  |
|            |  |  |            |         |         |        |  |        |  |        |  |
|            |  |  | Pilier     |         |         |        |  |        |  |        |  |
|            |  |  |            |         |         |        |  |        |  |        |  |
|            |  |  |            |         |         |        |  |        |  |        |  |
|            |  |  |            | Poisson |         |        |  |        |  |        |  |
|            |  |  |            |         |         |        |  |        |  |        |  |
|            |  |  |            |         |         |        |  | Pilier |  |        |  |
|            |  |  |            |         |         |        |  |        |  |        |  |
|            |  |  |            |         | Poisson |        |  |        |  |        |  |

Poisson mange près de (4,4)  
Bancs : 3 | Nourritures : 1

=== Etape 2 ===

|            |  |        |         |         |  |        |  |        |        |  |
|------------|--|--------|---------|---------|--|--------|--|--------|--------|--|
| Nourriture |  |        |         |         |  |        |  |        |        |  |
|            |  |        |         |         |  |        |  |        | Requin |  |
|            |  |        |         |         |  |        |  |        |        |  |
|            |  |        |         |         |  | Requin |  |        |        |  |
|            |  |        |         |         |  |        |  |        |        |  |
|            |  | Pilier |         |         |  |        |  |        |        |  |
|            |  |        | Poisson |         |  |        |  |        |        |  |
|            |  |        |         |         |  |        |  |        |        |  |
|            |  |        |         | Poisson |  |        |  | Pilier |        |  |
|            |  |        |         |         |  |        |  |        |        |  |

Bancs : 3 | Nourritures : 1

=== Etape 3 ===

|            |  |         |         |  |  |        |        |  |        |  |
|------------|--|---------|---------|--|--|--------|--------|--|--------|--|
| Nourriture |  |         |         |  |  |        |        |  |        |  |
|            |  |         |         |  |  |        |        |  | Requin |  |
|            |  | Poisson |         |  |  |        |        |  |        |  |
|            |  |         |         |  |  | Requin |        |  |        |  |
|            |  |         |         |  |  |        |        |  |        |  |
|            |  | Pilier  | Poisson |  |  |        |        |  |        |  |
|            |  |         |         |  |  |        |        |  |        |  |
|            |  |         | Poisson |  |  |        |        |  |        |  |
|            |  |         |         |  |  |        | Pilier |  |        |  |
|            |  |         |         |  |  |        |        |  |        |  |

Poisson mange près de (1,1)

Bancs : 3 | Nourritures : 1

=== Etape 4 ===

|            |         |         |  |  |  |        |  |        |        |  |  |
|------------|---------|---------|--|--|--|--------|--|--------|--------|--|--|
| Nourriture |         |         |  |  |  |        |  |        |        |  |  |
|            | Poisson |         |  |  |  |        |  |        | Requin |  |  |
|            |         |         |  |  |  |        |  |        |        |  |  |
|            |         |         |  |  |  | Requin |  |        |        |  |  |
|            |         | Poisson |  |  |  |        |  |        |        |  |  |
|            |         | Pilier  |  |  |  |        |  |        |        |  |  |
|            |         | Poisson |  |  |  |        |  |        |        |  |  |
|            |         |         |  |  |  |        |  |        |        |  |  |
|            |         |         |  |  |  |        |  | Pilier |        |  |  |
|            |         |         |  |  |  |        |  |        |        |  |  |

Poisson mange près de (1,1)

Bancs : 3 | Nourritures : 0

=== Etape 5 ===

|  |         |        |  |  |  |        |  |        |        |  |  |
|--|---------|--------|--|--|--|--------|--|--------|--------|--|--|
|  |         |        |  |  |  |        |  |        |        |  |  |
|  |         |        |  |  |  |        |  |        | Requin |  |  |
|  |         |        |  |  |  |        |  |        |        |  |  |
|  | Poisson |        |  |  |  | Requin |  |        |        |  |  |
|  |         |        |  |  |  |        |  |        |        |  |  |
|  | Poisson | Pilier |  |  |  |        |  |        |        |  |  |
|  |         |        |  |  |  |        |  |        |        |  |  |
|  |         |        |  |  |  |        |  |        |        |  |  |
|  |         |        |  |  |  |        |  | Pilier |        |  |  |
|  |         |        |  |  |  |        |  |        |        |  |  |

Bancs : 3 | Nourritures : 0

=== Etape 6 ===

[illegible]

Bancs : 3 | Nourritures : 0

### === Etape 7 ===

[illegible]

Bancs : 3 | Nourritures : 0

=== Etape 8 ===

|  |         |         |  |  |  |        |  |        |  |        |
|--|---------|---------|--|--|--|--------|--|--------|--|--------|
|  |         |         |  |  |  |        |  |        |  |        |
|  | Poisson |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  | Poisson |         |  |  |  |        |  |        |  | Requin |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  | Requin |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         | Pilier  |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         | Poisson |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  | Pilier |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |
|  |         |         |  |  |  |        |  |        |  |        |

Bancs : 3 | Nourritures : 0

=== Etape 9 ===

|  |         |        |  |  |  |        |  |        |  |        |
|--|---------|--------|--|--|--|--------|--|--------|--|--------|
|  |         |        |  |  |  |        |  |        |  |        |
|  | Poisson |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  | Poisson |        |  |  |  |        |  |        |  | Requin |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  | Requin |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  | Poisson | Pilier |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  | Pilier |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |
|  |         |        |  |  |  |        |  |        |  |        |

Bancs : 3 | Nourritures : 0

=== Etape 10 ===

|  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |  |













The image shows a musical score for a piece titled "Poisson" and "Pilier". The score is written on a grand staff with two systems. The first system has a treble clef and a key signature of one flat (B-flat). The second system has a bass clef and a key signature of one flat (B-flat). The music consists of a single melodic line with various note values and rests. The title "Poisson" is written above the first system, and "Pilier" is written above the second system.

=== Etape 21 ===

[illegible]

=== Etape 22 ===

[illegible]

|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|  
 |-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|  
 |Poisson |-----|-----|-----|-----|-----|-----|-----|-----|Pilier |-----|  
 |-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|  
 |-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

=== Etape 23 ===

[illegible]

=== Etape 24 ===

[illegible]

Bancs : 3 | Nourritures : 0

=== Etape 25 ===

[illegible]

=== Etape 26 ===

[illegible]

```

:-----:
|      |      |      |      |      |      |      |      |
:-----:

```

[REQUIN] mange un poisson du banc en (-1,-1)  
 Bancs : 2 | Nourritures : 0

=== Etape 27 ===

```

:-----:
|Poisson |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |Requin |
:-----:
|      |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |Requin |      |      |
:-----:
|      |      |      |      |      |      |      |      |
:-----:
|      |      |Pilier |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |Pilier |
:-----:
|Poisson |      |      |      |      |      |      |      |
:-----:

```

Bancs : 1 | Nourritures : 0

=== Etape 28 ===

```

:-----:
|      |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |Requin |
:-----:
|      |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |Requin |      |      |
:-----:
|      |      |      |      |      |      |      |      |
:-----:
|      |      |Pilier |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |Pilier |
:-----:

```



Bancs : 1 | Nourritures : 0

=== Nombre max d'étapes atteint ===

PS C:\Users\Imad\Documents\GitHub\Cours-Sorbonne-Imad\S4\Java\Projet  
Java\UL2IN002-PROJET\codeImad\Nouveau dossier (2)>java TestSimulation 0  
Scénario généré : 2 bancs, 3 nourritures, 2 requins, 3 piliers

=== Etape 1 ===

```
:-----:
|      |      |      |Requin |      |      |      |      |      |
:-----:
|      |      |      |Nourriture|      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |Pilier |      |
:-----:
|Requin |      |      |      |      |      |      |      |      |
:-----:
|      |Nourriture|      |      |      |      |      |      |      |
:-----:
|      |      |      |Poisson |      |      |Pilier |      |      |
:-----:
|      |Pilier |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |Poisson |      |
:-----:
|      |      |      |      |      |      |      |Nourriture|      |
:-----:
```

Poisson mange près de (10,8)

Poisson mange près de (5,2)

Bancs : 2 | Nourritures : 3

=== Etape 2 ===

```
:-----:
|      |      |      |Requin |      |      |      |      |      |
:-----:
|      |      |      |Nourriture|      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |Pilier |      |
:-----:
|Requin |      |      |      |      |      |      |      |      |
:-----:
|      |Nourriture|Poisson |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |Pilier |      |      |
:-----:
|      |Pilier |      |      |      |      |      |      |      |
:-----:
|      |      |      |      |      |      |      |      |      |
:-----:
```





|  |  |  |  |  |  |         |  |  |  |  |
|--|--|--|--|--|--|---------|--|--|--|--|
|  |  |  |  |  |  | Poisson |  |  |  |  |
|  |  |  |  |  |  |         |  |  |  |  |
|  |  |  |  |  |  |         |  |  |  |  |
|  |  |  |  |  |  |         |  |  |  |  |

[REQUIN] mange un poisson du banc en (4,2)

Poisson mange près de (2,4)

Bancs : 2 | Nourritures : 3

=== Etape 5 ===

|        |        |         |            |  |         |        |  |        |  |  |
|--------|--------|---------|------------|--|---------|--------|--|--------|--|--|
|        |        |         | Requin     |  |         |        |  |        |  |  |
|        |        |         | Nourriture |  |         |        |  |        |  |  |
|        |        | Poisson |            |  |         |        |  | Pilier |  |  |
| Requin |        |         |            |  |         |        |  |        |  |  |
|        |        |         |            |  |         |        |  |        |  |  |
|        |        |         |            |  |         | Pilier |  |        |  |  |
|        | Pilier |         |            |  | Poisson |        |  |        |  |  |
|        |        |         |            |  |         |        |  |        |  |  |
|        |        |         |            |  |         |        |  |        |  |  |
|        |        |         |            |  |         |        |  |        |  |  |
|        |        |         |            |  |         |        |  |        |  |  |

[REQUIN] mange un poisson du banc en (2,4)

Poisson mange près de (2,4)

Bancs : 2 | Nourritures : 3

=== Etape 6 ===

|        |  |  |            |         |  |        |  |        |  |  |
|--------|--|--|------------|---------|--|--------|--|--------|--|--|
|        |  |  | Requin     |         |  |        |  |        |  |  |
|        |  |  | Nourriture |         |  |        |  |        |  |  |
|        |  |  |            |         |  |        |  | Pilier |  |  |
| Requin |  |  |            |         |  |        |  |        |  |  |
|        |  |  |            |         |  |        |  |        |  |  |
|        |  |  |            |         |  |        |  |        |  |  |
|        |  |  |            | Poisson |  | Pilier |  |        |  |  |

|  |        |  |  |  |  |  |  |  |  |  |
|--|--------|--|--|--|--|--|--|--|--|--|
|  |        |  |  |  |  |  |  |  |  |  |
|  | Pilier |  |  |  |  |  |  |  |  |  |
|  |        |  |  |  |  |  |  |  |  |  |
|  |        |  |  |  |  |  |  |  |  |  |
|  |        |  |  |  |  |  |  |  |  |  |
|  |        |  |  |  |  |  |  |  |  |  |
|  |        |  |  |  |  |  |  |  |  |  |
|  |        |  |  |  |  |  |  |  |  |  |
|  |        |  |  |  |  |  |  |  |  |  |

[REQUIN] mange un poisson du banc en (2,4)

[REQUIN] mange un poisson du banc en (-1,-1)

Bancs : 1 | Nourritures : 3

=== Etape 7 ===

|        |        |  |         |  |  |  |        |  |        |  |
|--------|--------|--|---------|--|--|--|--------|--|--------|--|
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  | Requin  |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  | Pilier |  |
|        |        |  |         |  |  |  |        |  |        |  |
| Requin |        |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  | Poisson |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  | Pilier |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        | Pilier |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |
|        |        |  |         |  |  |  |        |  |        |  |

Bancs : 1 | Nourritures : 3

=== Etape 8 ===

|        |  |  |         |  |  |  |        |  |        |  |
|--------|--|--|---------|--|--|--|--------|--|--------|--|
|        |  |  |         |  |  |  |        |  |        |  |
|        |  |  | Requin  |  |  |  |        |  |        |  |
|        |  |  |         |  |  |  |        |  |        |  |
|        |  |  |         |  |  |  |        |  |        |  |
|        |  |  |         |  |  |  |        |  |        |  |
|        |  |  |         |  |  |  |        |  | Pilier |  |
|        |  |  |         |  |  |  |        |  |        |  |
| Requin |  |  | Poisson |  |  |  |        |  |        |  |
|        |  |  |         |  |  |  |        |  |        |  |
|        |  |  |         |  |  |  |        |  |        |  |
|        |  |  |         |  |  |  |        |  |        |  |
|        |  |  |         |  |  |  | Pilier |  |        |  |

[illegible]

[REQUIN] mange un poisson du banc en (3,2)

Bancs : 1 | Nourritures : 3

=== Etape 9 ===

[illegible]

Bancs : 1 | Nourritures : 3

=== Etape 10 ===

[illegible]







| | | | | | | | | | | |  
-----  
Bancs : 1 | Nourritures : 3

=== Etape 17 ===

|Poisson | | |Requin | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----  
| | | | | | | | |Pilier | | |  
-----  
|Requin | | | | | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----  
| | | | | | |Pilier | | | | | |  
-----  
| |Pilier | | | | | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----

Bancs : 1 | Nourritures : 3

=== Etape 18 ===

| |Poisson | |Requin | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----  
| | | | | | | | |Pilier | | |  
-----  
|Requin | | | | | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----  
| | | | | | |Pilier | | | | | |  
-----  
| |Pilier | | | | | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----  
| | | | | | | | | | | |  
-----



Bancs : 1 | Nourritures : 3

=== Etape 19 ===

|         |        |  |        |  |  |  |        |        |  |  |  |
|---------|--------|--|--------|--|--|--|--------|--------|--|--|--|
| Poisson |        |  | Requin |  |  |  |        |        |  |  |  |
|         |        |  |        |  |  |  |        |        |  |  |  |
|         |        |  |        |  |  |  |        | Pilier |  |  |  |
| Requin  |        |  |        |  |  |  |        |        |  |  |  |
|         |        |  |        |  |  |  |        |        |  |  |  |
|         |        |  |        |  |  |  | Pilier |        |  |  |  |
|         | Pilier |  |        |  |  |  |        |        |  |  |  |
|         |        |  |        |  |  |  |        |        |  |  |  |
|         |        |  |        |  |  |  |        |        |  |  |  |
|         |        |  |        |  |  |  |        |        |  |  |  |
|         |        |  |        |  |  |  |        |        |  |  |  |

Bancs : 1 | Nourritures : 3

=== Etape 20 ===

|        |         |  |        |  |  |  |        |        |  |  |  |
|--------|---------|--|--------|--|--|--|--------|--------|--|--|--|
|        | Poisson |  | Requin |  |  |  |        |        |  |  |  |
|        |         |  |        |  |  |  |        |        |  |  |  |
|        |         |  |        |  |  |  |        | Pilier |  |  |  |
| Requin |         |  |        |  |  |  |        |        |  |  |  |
|        |         |  |        |  |  |  |        |        |  |  |  |
|        |         |  |        |  |  |  | Pilier |        |  |  |  |
|        | Pilier  |  |        |  |  |  |        |        |  |  |  |
|        |         |  |        |  |  |  |        |        |  |  |  |
|        |         |  |        |  |  |  |        |        |  |  |  |
|        |         |  |        |  |  |  |        |        |  |  |  |
|        |         |  |        |  |  |  |        |        |  |  |  |

Bancs : 1 | Nourritures : 3

[illegible][illegible]

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Requin | Pilier | | | | | | | | | |  
Bancs : 3 | Nourritures : 3

=== Etape 22 ===

[illegible]

=== Etape 23 ===

|         |         |        |        |  |  |  |  |  |  |
|---------|---------|--------|--------|--|--|--|--|--|--|
| Poisson | Pilier  |        |        |  |  |  |  |  |  |
| Poisson | Poisson |        |        |  |  |  |  |  |  |
|         |         | Pilier |        |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |
|         |         | Pilier |        |  |  |  |  |  |  |
|         | Requin  |        |        |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |
| Pilier  |         |        | Pilier |  |  |  |  |  |  |
|         | Requin  |        | Pilier |  |  |  |  |  |  |

Bancs : 3 | Nourritures : 3

=== Etape 24 ===

|         |         |        |        |  |  |  |  |  |  |  |
|---------|---------|--------|--------|--|--|--|--|--|--|--|
|         | Poisson | Pilier |        |  |  |  |  |  |  |  |
| Poisson | Poisson |        |        |  |  |  |  |  |  |  |
|         |         | Pilier |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         | Pilier |        |  |  |  |  |  |  |  |
|         | Requin  |        |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
| Pilier  |         |        | Pilier |  |  |  |  |  |  |  |
|         | Requin  | Pilier |        |  |  |  |  |  |  |  |

Bancs : 3 | Nourritures : 3

=== Etape 25 ===

|         |         |        |        |  |  |  |  |  |  |  |
|---------|---------|--------|--------|--|--|--|--|--|--|--|
| Poisson | Poisson | Pilier |        |  |  |  |  |  |  |  |
|         | Poisson |        |        |  |  |  |  |  |  |  |
|         |         | Pilier |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         | Pilier |        |  |  |  |  |  |  |  |
|         | Requin  |        |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
| Pilier  |         |        | Pilier |  |  |  |  |  |  |  |
|         | Requin  | Pilier |        |  |  |  |  |  |  |  |

Bancs : 3 | Nourritures : 3

[illegible]

=== Etape 27 ===

[illegible]

=== Etape 28 ===

\_\_\_\_\_

|         |         |        |        |  |  |  |  |  |  |  |
|---------|---------|--------|--------|--|--|--|--|--|--|--|
| Poisson | Poisson | Pilier |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         | Pilier |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         | Pilier |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         | Requin  |        |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
| Pilier  |         |        | Pilier |  |  |  |  |  |  |  |
|         |         |        |        |  |  |  |  |  |  |  |
|         | Requin  |        | Pilier |  |  |  |  |  |  |  |

Bancs : 1 | Nourritures : 3

=== Etape 29 ===

|         |        |        |        |  |  |  |  |  |  |  |
|---------|--------|--------|--------|--|--|--|--|--|--|--|
|         |        | Pilier |        |  |  |  |  |  |  |  |
|         |        |        |        |  |  |  |  |  |  |  |
| Poisson |        |        |        |  |  |  |  |  |  |  |
|         |        | Pilier |        |  |  |  |  |  |  |  |
|         |        |        |        |  |  |  |  |  |  |  |
|         |        |        |        |  |  |  |  |  |  |  |
|         |        |        |        |  |  |  |  |  |  |  |
|         |        | Pilier |        |  |  |  |  |  |  |  |
|         |        |        |        |  |  |  |  |  |  |  |
|         | Requin |        |        |  |  |  |  |  |  |  |
|         |        |        |        |  |  |  |  |  |  |  |
|         |        |        |        |  |  |  |  |  |  |  |
| Pilier  |        |        | Pilier |  |  |  |  |  |  |  |
|         |        |        |        |  |  |  |  |  |  |  |
|         | Requin |        | Pilier |  |  |  |  |  |  |  |

Bancs : 1 | Nourritures : 3

=== Etape 30 ===

|         |  |        |  |  |  |  |  |  |  |  |
|---------|--|--------|--|--|--|--|--|--|--|--|
|         |  |        |  |  |  |  |  |  |  |  |
| Poisson |  | Pilier |  |  |  |  |  |  |  |  |

|        |        |   |        |        |   |   |   |   |   |   |
|--------|--------|---|--------|--------|---|---|---|---|---|---|
|        |        |   |        |        |   |   |   |   |   |   |
| :      | :      | : | :      | :      | : | : | : | : | : | : |
|        |        |   | Pilier |        |   |   |   |   |   |   |
| :      | :      | : | :      | :      | : | : | : | : | : | : |
|        |        |   |        |        |   |   |   |   |   |   |
| :      | :      | : | :      | :      | : | : | : | : | : | : |
|        |        |   |        |        |   |   |   |   |   |   |
| :      | :      | : | :      | :      | : | : | : | : | : | : |
|        |        |   | Pilier |        |   |   |   |   |   |   |
| :      | :      | : | :      | :      | : | : | : | : | : | : |
|        | Requin |   |        |        |   |   |   |   |   |   |
| :      | :      | : | :      | :      | : | : | : | : | : | : |
|        |        |   |        |        |   |   |   |   |   |   |
| :      | :      | : | :      | :      | : | : | : | : | : | : |
| Pilier |        |   |        | Pilier |   |   |   |   |   |   |
| :      | :      | : | :      | :      | : | : | : | : | : | : |
|        | Requin |   | Pilier |        |   |   |   |   |   |   |
| :      | :      | : | :      | :      | : | : | : | : | : | : |

Bancs : 0 | Nourritures : 3

=== Simulation terminée ===

Copier-coller vos classes et interfaces à partir d'ici :

BancPoissons.java  
 EntiteMarine.java  
 ExceptionManager.java  
 NourritureManager.java  
 Pilier.java  
 PoissonManager.java  
 Requin.java  
 RequinManager.java  
 ScenarioGenerator.java  
 Simulation.java  
 SimulationException.java  
 TestSimulation.java